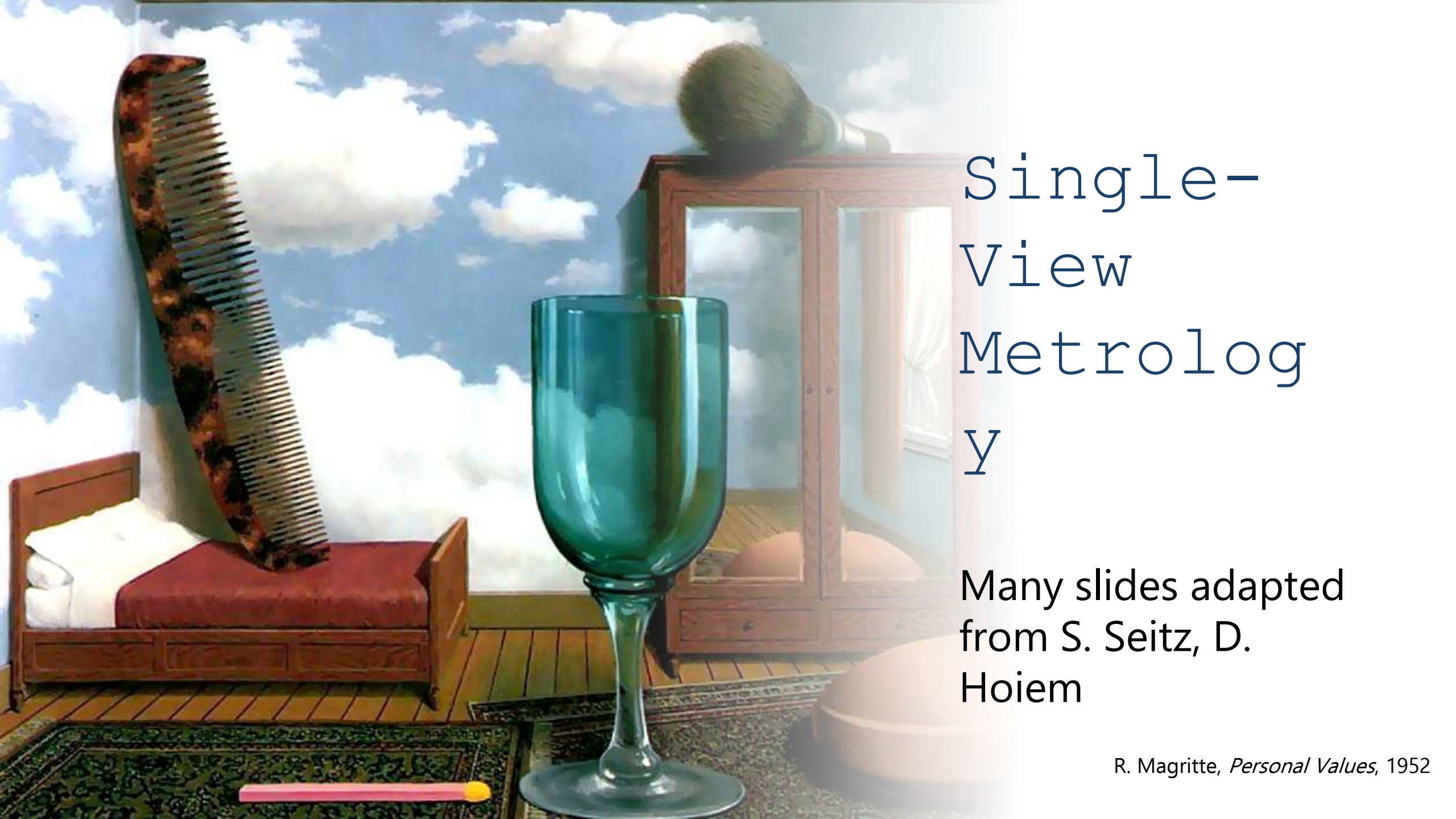




Single- View Metrolog y

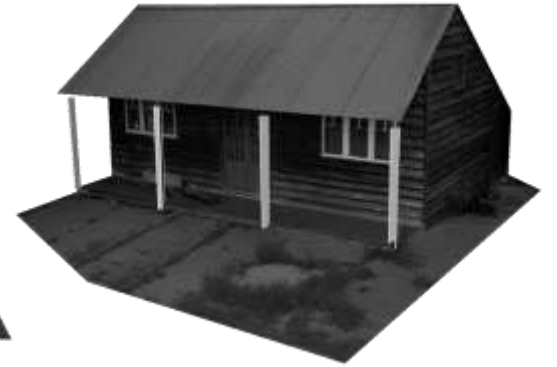
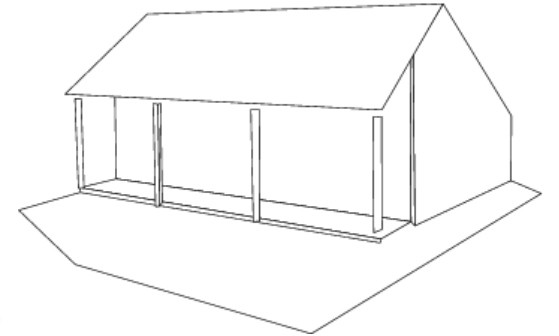
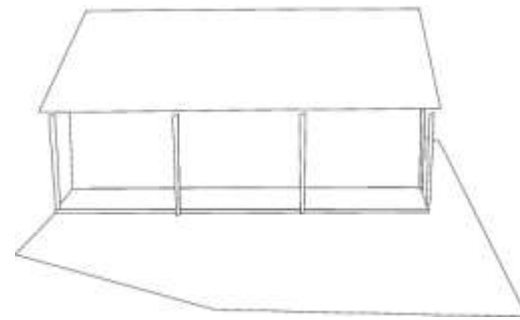
Many slides adapted
from S. Seitz, D.
Hoiem

Outline

- Camera calibration using vanishing points
- Measurements from a single image
- Applications of single-view metrology

Camera converts $3D \rightarrow 2D$
but no way to find height.

Application: 3D from a Single Image



Application: Image Editing, Augmented Reality



K. Karsch and V. Hedau and D. Forsyth and D. Hoiem. [Rendering Synthetic Objects into Legacy Photographs](#), SIGGRAPH Asia 2011

Camera Calibration Revisited

- What if world coordinates of reference 3D points are not known?
- We can use scene features such as vanishing points



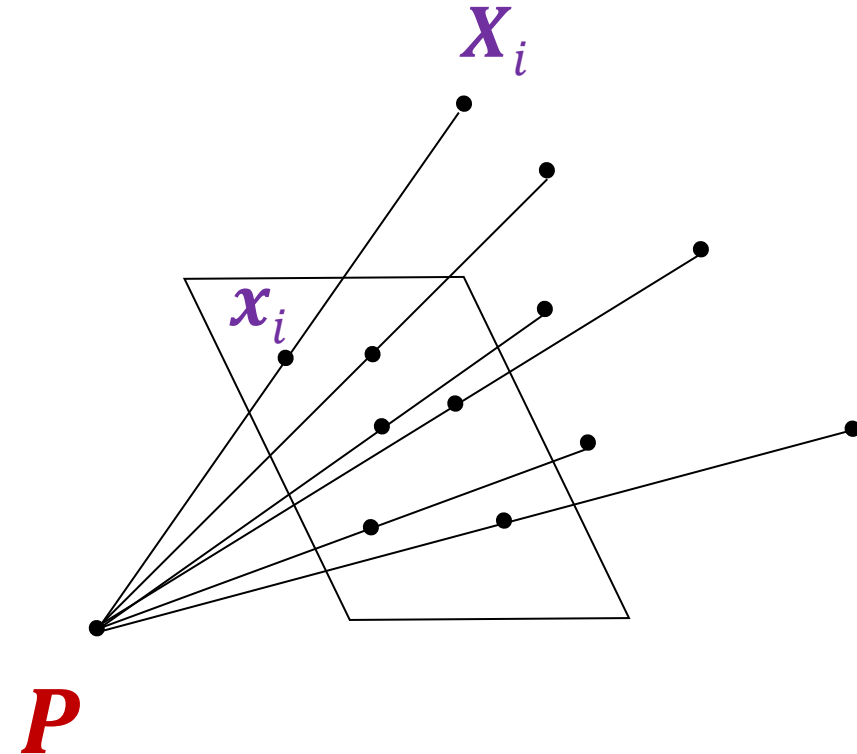
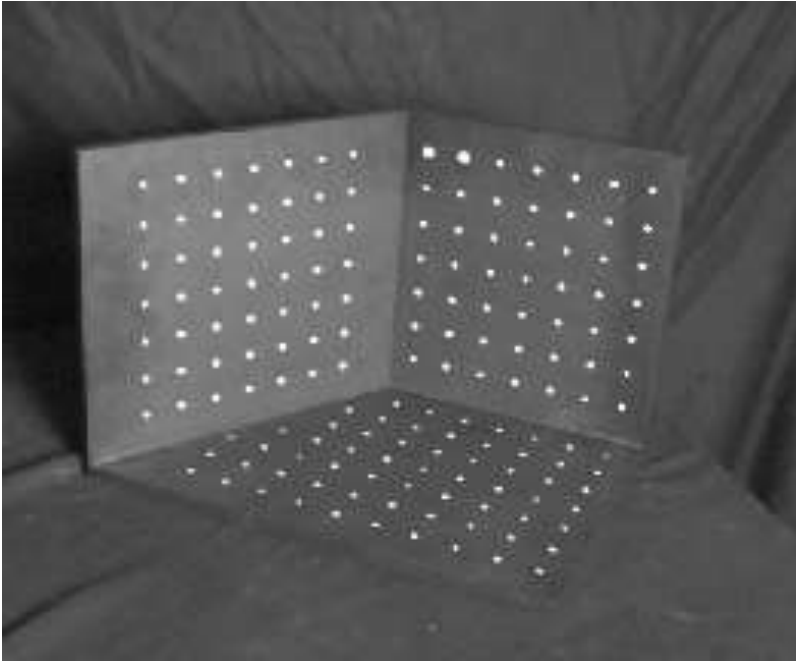
Last Time: Camera Calibration

$$\lambda \mathbf{x} = \mathbf{K}[\mathbf{R} \ \mathbf{t}]\mathbf{X}$$

$$\begin{pmatrix} \lambda x \\ \lambda y \\ \lambda \end{pmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

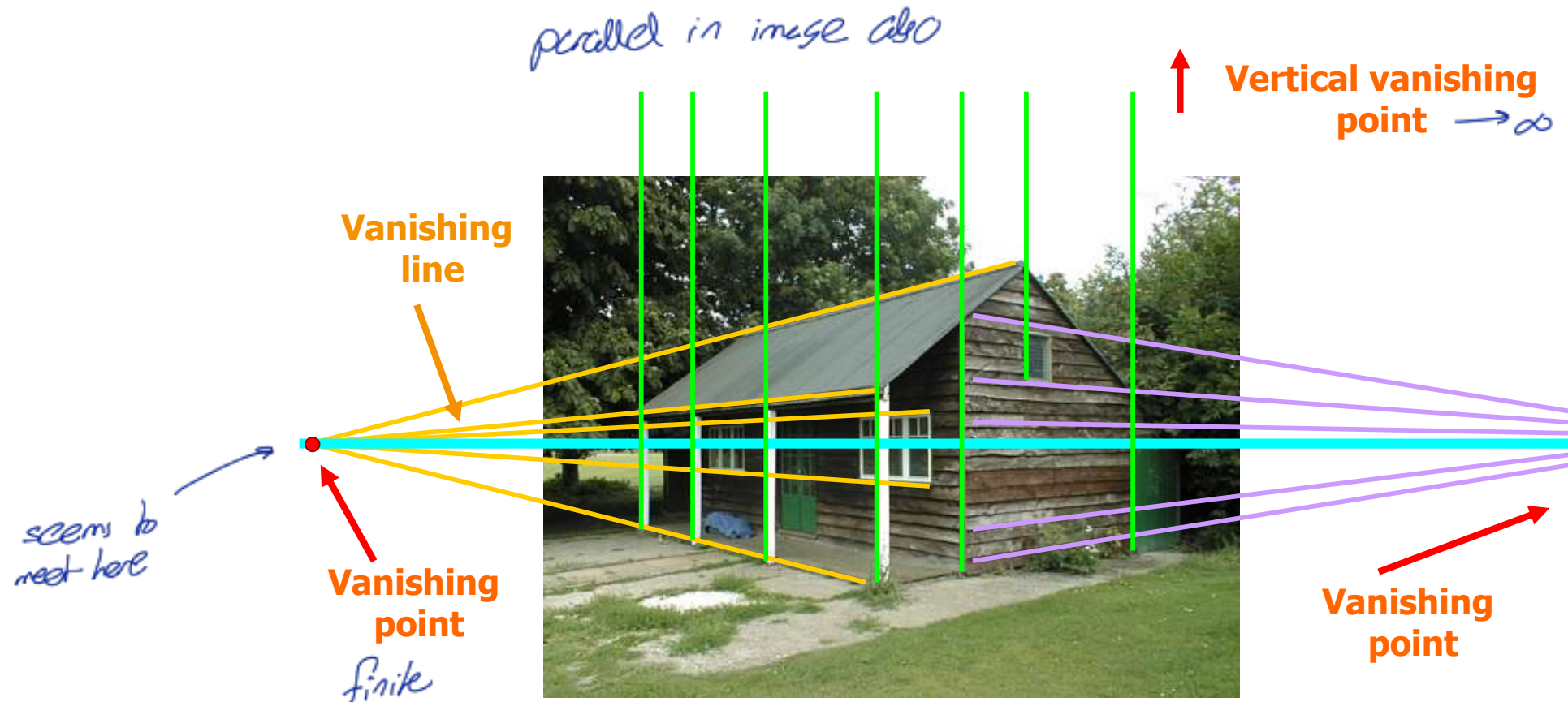
Last Time: Camera Calibration

- Given n points with known 3D coordinates $\mathbf{X}_i$ and known image projections $\mathbf{x}_i$, estimate the camera parameters

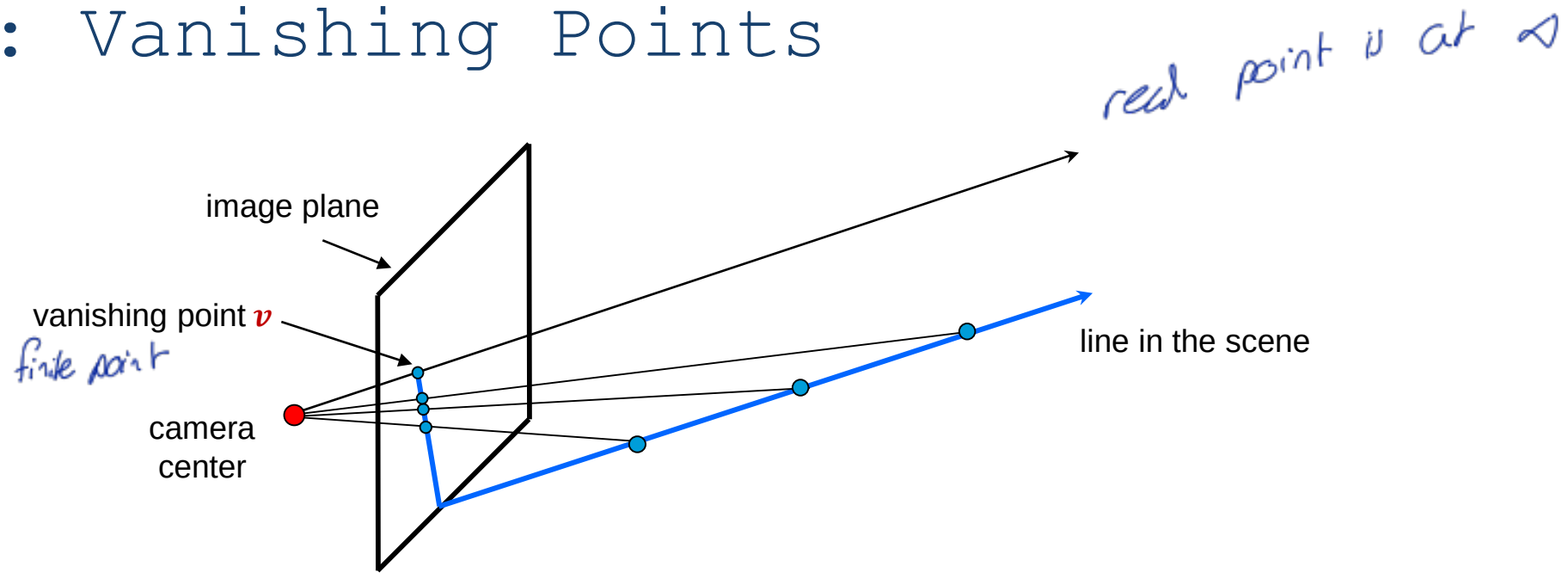


Camera Calibration Using Vanishing Points

- If world coordinates of reference 3D points are not known, in special cases, we may be able to use vanishing points



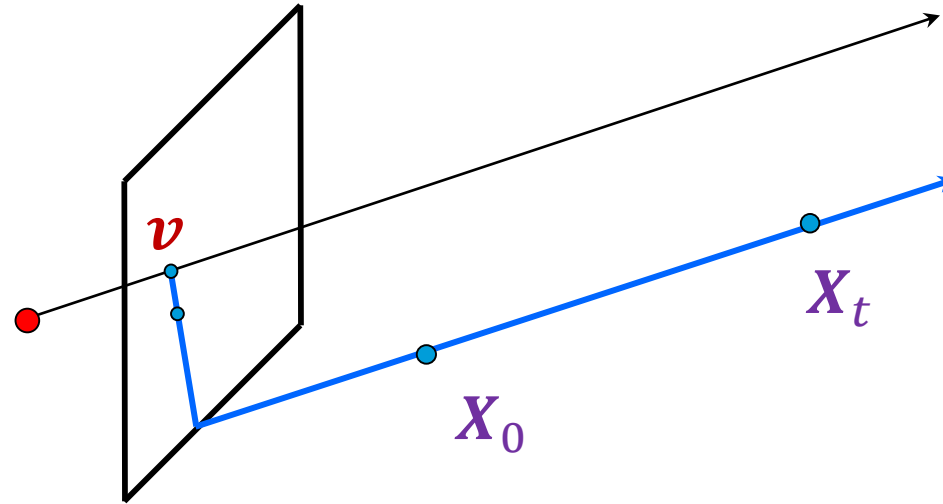
Review: Vanishing Points



- All lines having the same direction share the same vanishing point

parallel lines meet at the horizon $\Rightarrow$ railway

Computing Vanishing Points



- Let's parameterize the line using point $\mathbf{X}_0 = (X_0, Y_0, Z_0, 1)^T$ and direction vector $\mathbf{D} = (D_1, D_2, D_3)^T$:

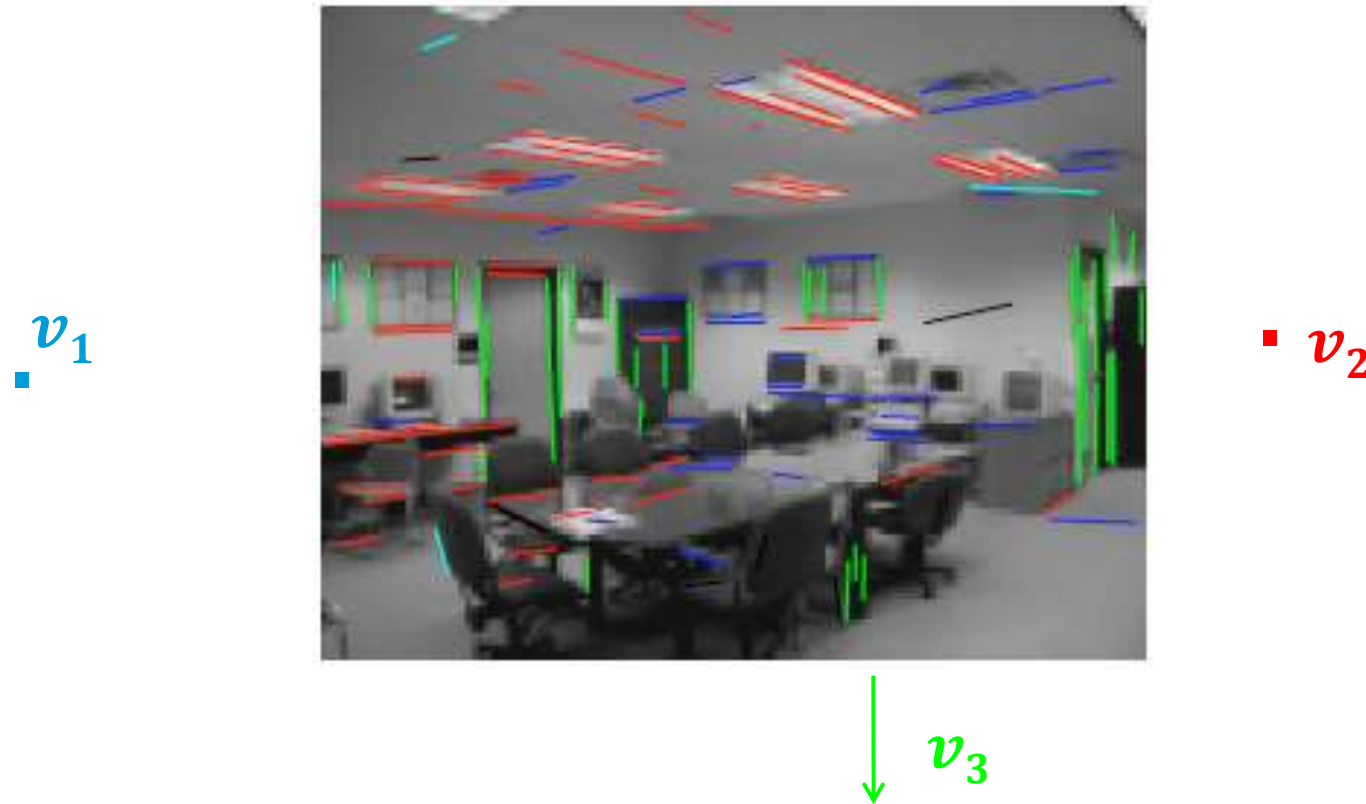
$$\mathbf{X}_t = \begin{pmatrix} X_0 + tD_1 \\ Y_0 + tD_2 \\ Z_0 + tD_3 \\ 1 \end{pmatrix} \equiv \begin{pmatrix} X_0/t + D_1 \\ Y_0/t + D_2 \\ Z_0/t + D_3 \\ 1/t \end{pmatrix} \quad \lim_{t \rightarrow \infty} \mathbf{X}_\infty = \begin{pmatrix} D_1 \\ D_2 \\ D_3 \\ 0 \end{pmatrix}$$

so point at ∞
is also clear that
why this is homogeneous.
when last coord is
0, it gives the
direction

- $\mathbf{X}_\infty$ is a point at infinity, $\mathbf{v}$ is its projection: $\mathbf{v} \equiv \mathbf{P}\mathbf{X}_\infty$

Calibration from Vanishing Points

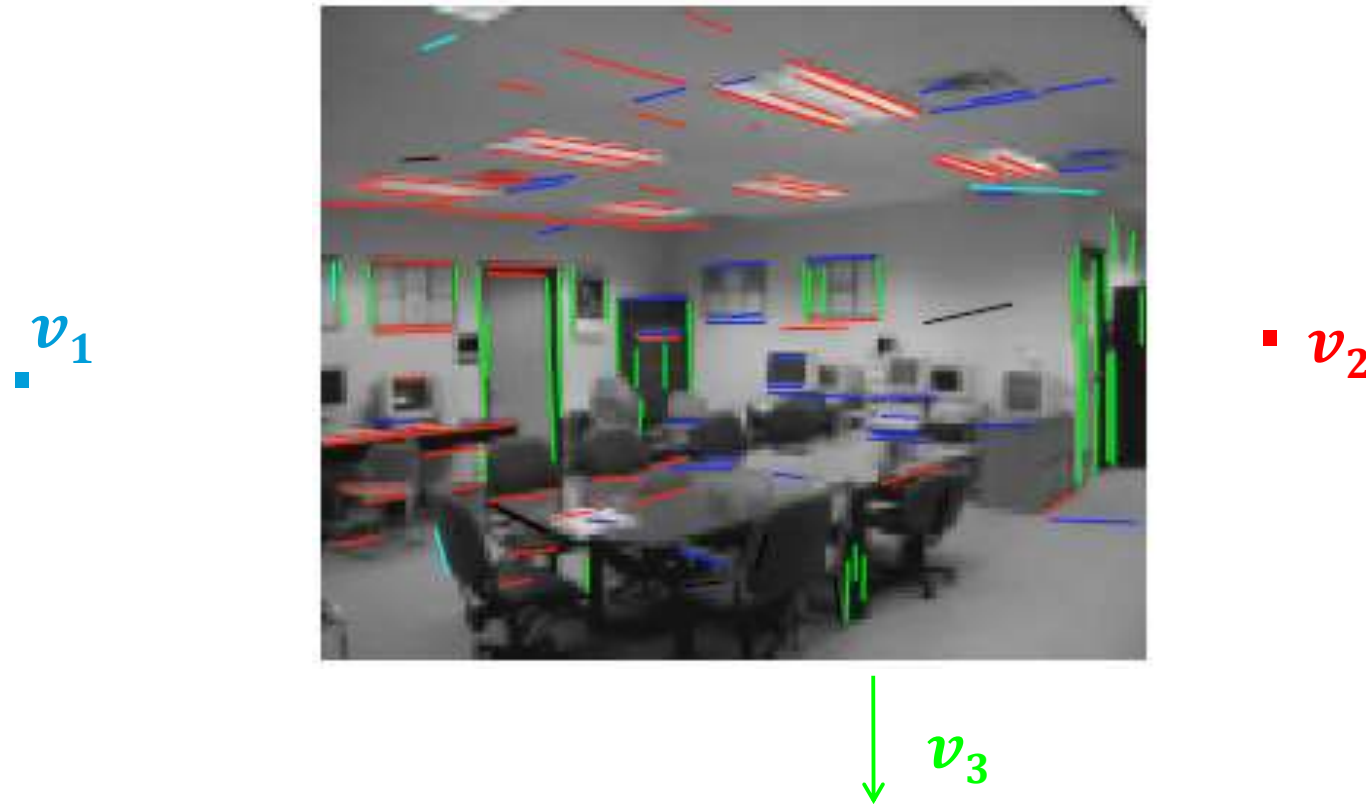
- Consider a scene with three orthogonal vanishing directions:



- Note: v_1 , v_2 are *finite* vanishing points and v_3 is an *infinite* vanishing point

Calibration from Vanishing Points

- Consider a scene with three orthogonal vanishing directions:



- We can align the world coordinate system with these directions

Calibration from Vanishing Points

$$\mathbf{P} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix}$$

$$\mathbf{p}_1 \quad \mathbf{p}_2 \quad \mathbf{p}_3 \quad \mathbf{p}_4$$

Calibration from Vanishing Points

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \mathbf{p}_1$$

$\mathbf{p}_1 \quad \mathbf{p}_2 \quad \mathbf{p}_3 \quad \mathbf{p}_4$

vanishing point in x-direction

Calibration from Vanishing Points

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \mathbf{p}_2 \quad \leftarrow \text{v in } g\text{-direction}$$

$\mathbf{p}_1 \quad \mathbf{p}_2 \quad \mathbf{p}_3 \quad \mathbf{p}_4$

Calibration from Vanishing Points

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \mathbf{p}_3$$

$$\underbrace{\mathbf{p}_1 \quad \mathbf{p}_2 \quad \mathbf{p}_3}_{\text{Vanishing points in } x, y, z \text{ directions (i.e., } \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)}$$

Vanishing points in
 x, y, z directions
(i.e., $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$)

Calibration from Vanishing Points

$$\begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \mathbf{p}_4$$

$\underbrace{\mathbf{p}_1 \quad \mathbf{p}_2 \quad \mathbf{p}_3}_{\text{Vanishing points in } x, y, z \text{ directions (i.e., } \mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)}$ $\mathbf{p}_4$ (Projection of world origin)

$\uparrow$ world origin

- Problem: this only gives us the four columns up to independent scale factors, additional constraints needed to solve for them

we don't know K

Calibration from Vanishing Points

- Let us align the world coordinate system with three orthogonal vanishing directions in the scene:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\mathbf{v}_i \equiv \mathbf{K}[\mathbf{R}|\mathbf{t}] \begin{pmatrix} \mathbf{e}_i \\ 0 \end{pmatrix}$$

vanish point in
x, y, z direction
as i change.

Calibration from Vanishing Points

- Let us align the world coordinate system with three orthogonal vanishing directions in the scene:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\mathbf{v}_i \equiv \mathbf{K} \mathbf{R} \mathbf{e}_i$$

$$\mathbf{e}_i \equiv \mathbf{R}^T \mathbf{K}^{-1} \mathbf{v}_i$$

- Orthogonality constraint: $\mathbf{e}_i^T \mathbf{e}_j = 0$

$$\underbrace{\mathbf{v}_i^T \mathbf{K}^{-T} \mathbf{R}}_{\mathbf{e}_i^T} \underbrace{\mathbf{R}^T \mathbf{K}^{-1} \mathbf{v}_j}_{\mathbf{e}_j} = 0$$

Calibration from Vanishing Points

- Let us align the world coordinate system with three orthogonal vanishing directions in the scene:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{e}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\mathbf{v}_i \equiv \mathbf{K} \mathbf{R} \mathbf{e}_i$$

$$\mathbf{e}_i \equiv \mathbf{R}^T \mathbf{K}^{-1} \mathbf{v}_i$$

- Orthogonality constraint: $\mathbf{e}_i^T \mathbf{e}_j = 0$

$$\mathbf{v}_i^T \mathbf{K}^{-T} \mathbf{K}^{-1} \mathbf{v}_j = 0$$

- Extrinsic parameter matrix ($\mathbf{R}$) disappears and we are left with constraints on the calibration matrix!

Calibration from Vanishing Points

$$\mathbf{v}_i^T \mathbf{K}^{-T} \mathbf{K}^{-1} \mathbf{v}_j = 0$$

- How many constraints do we get?
 - Three: one for each pair of vanishing points
- How many unknown parameters does $\mathbf{K}$ have?
 - Three: f, p_x, p_y
- A couple of complications:
 - The constraints are nonlinear, but it's not hard to do the algebra (omitted)
 - At least two *finite* vanishing points are needed to solve for both focal length and principal point

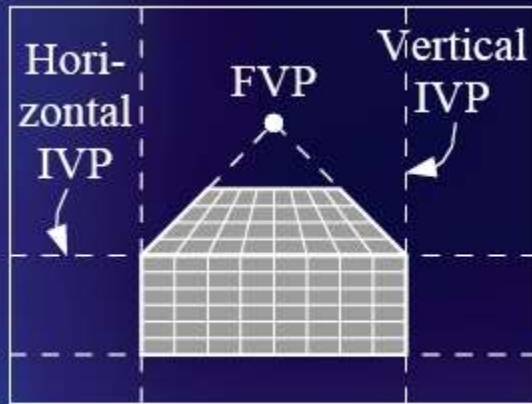
Calibration from vanishing points

bad

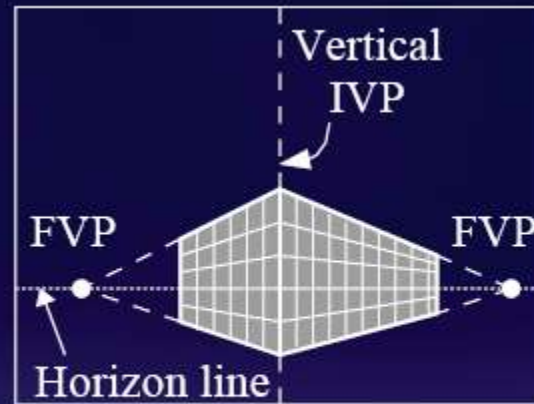
good

good

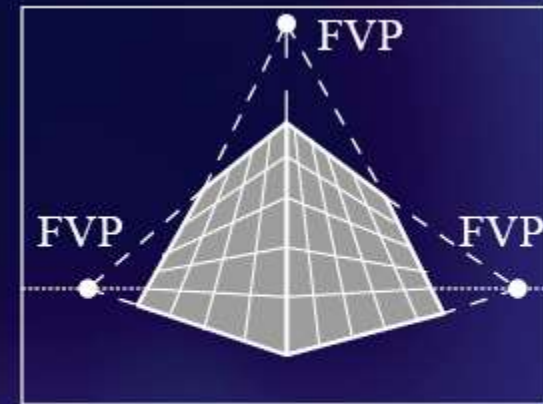
*good
can solve*



1 finite vanishing point,
2 infinite vanishing points



2 finite vanishing points,
1 infinite vanishing point



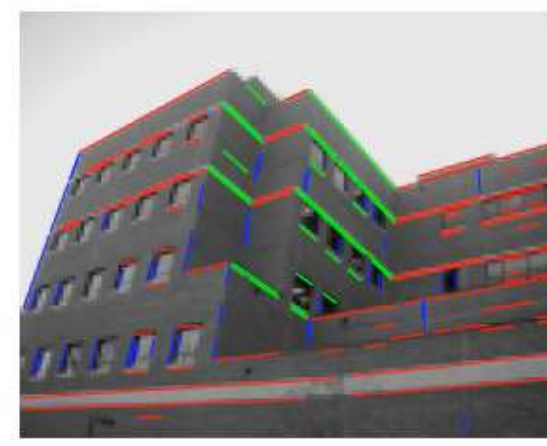
3 finite vanishing points



Cannot recover focal
length, principal point is
the third vanishing point



Can solve for focal length, principal point



Calibration from Vanishing Points

- Constraints on vanishing points: $\mathbf{v}_i \equiv \mathbf{K}\mathbf{R}\mathbf{e}_i$
- We just used orthogonality constraints to solve for $\mathbf{K}$
- Now we have:

$$\mathbf{K}^{-1}\mathbf{v}_i \equiv \mathbf{R}\mathbf{e}_i$$

$$\text{Notice: } \mathbf{R}\mathbf{e}_1 = [\mathbf{r}_1 \quad \mathbf{r}_2 \quad \mathbf{r}_3] \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \mathbf{r}_1$$

$$\text{Thus, } \mathbf{r}_i \equiv \mathbf{K}^{-1}\mathbf{v}_i$$

- The scale ambiguity goes away since we require $\|\mathbf{r}_i\|^2 = 1$

Calibration from Vanishing Points: Summary

1. Solve for intrinsic parameters (focal length, principal point) using three orthogonal vanishing points
 2. Get extrinsic parameters (rotation) directly from vanishing points once calibration matrix is known
- Advantages
 - No need for calibration chart, 2D-3D correspondences
 - Could be completely automatic
 - Disadvantages
 - Only applies to certain kinds of scenes
 - It is tricky to accurately localize vanishing points
 - Need at least two finite vanishing points

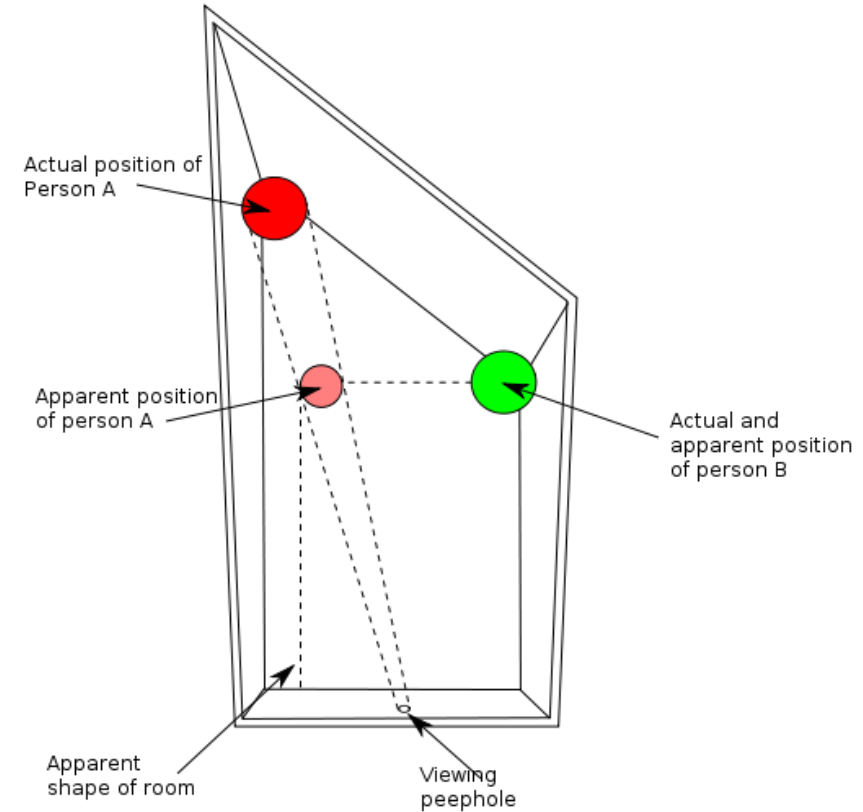
Outline

- Camera calibration using vanishing points
- Measurements from a single image
 - Measuring height
 - Measuring within planes



Making Measurements from a Single Image

- Reminder: Caveat emptor!

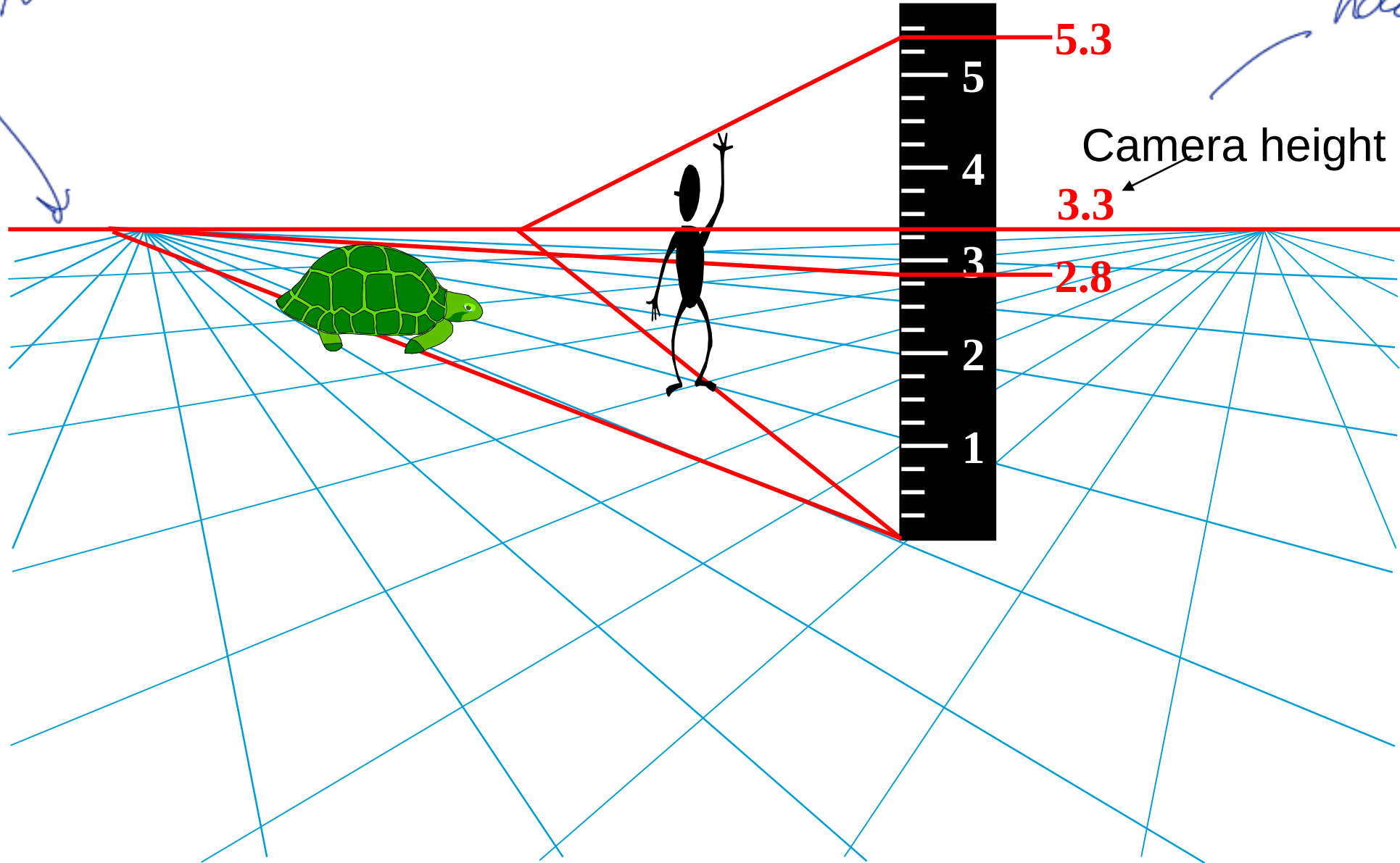


http://en.wikipedia.org/wiki/Ames_room

Recall: Measuring Height with a Ruler

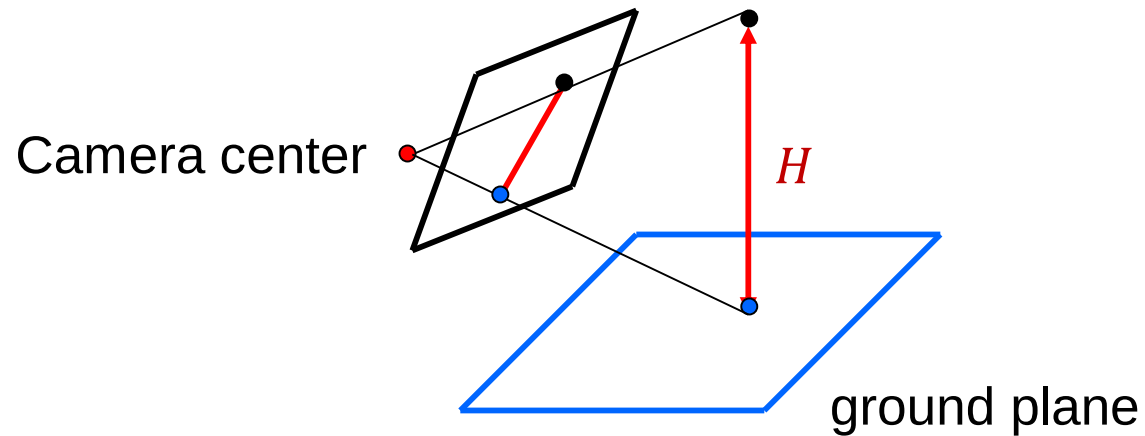
horizon

how??

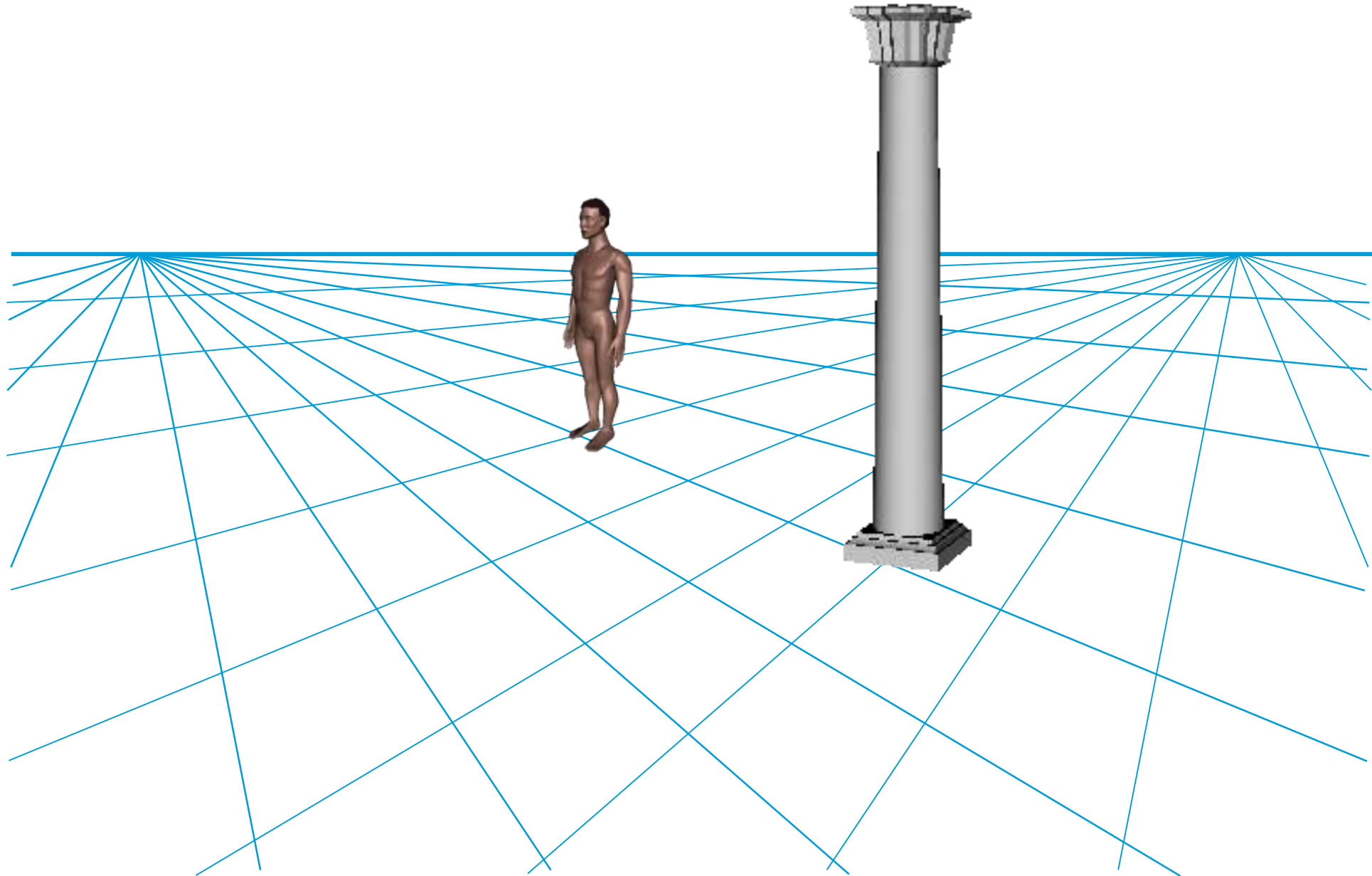


Measuring Height without a Ruler

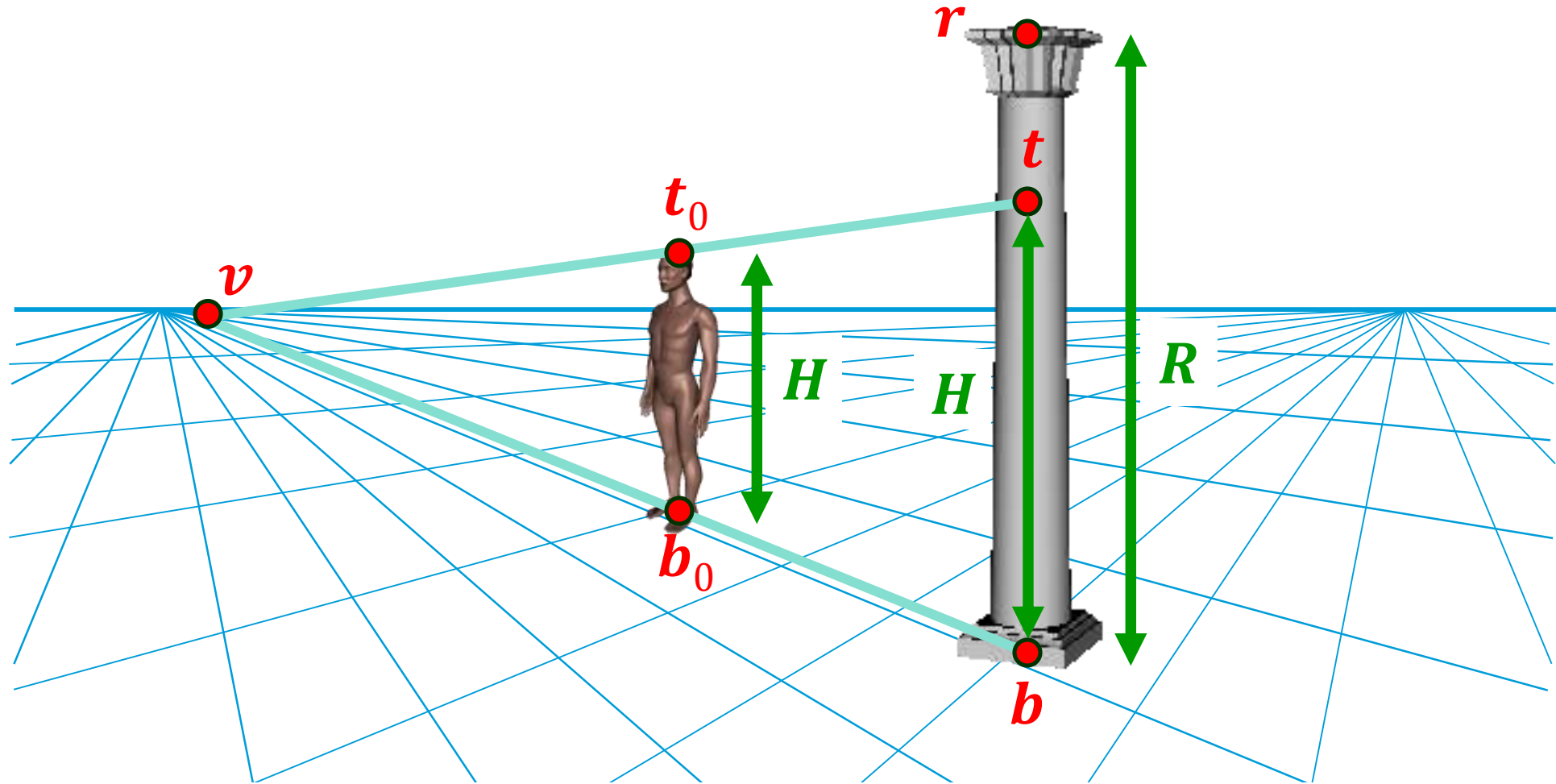
- Goal: compute height H from image measurements



Measuring Height without a Ruler

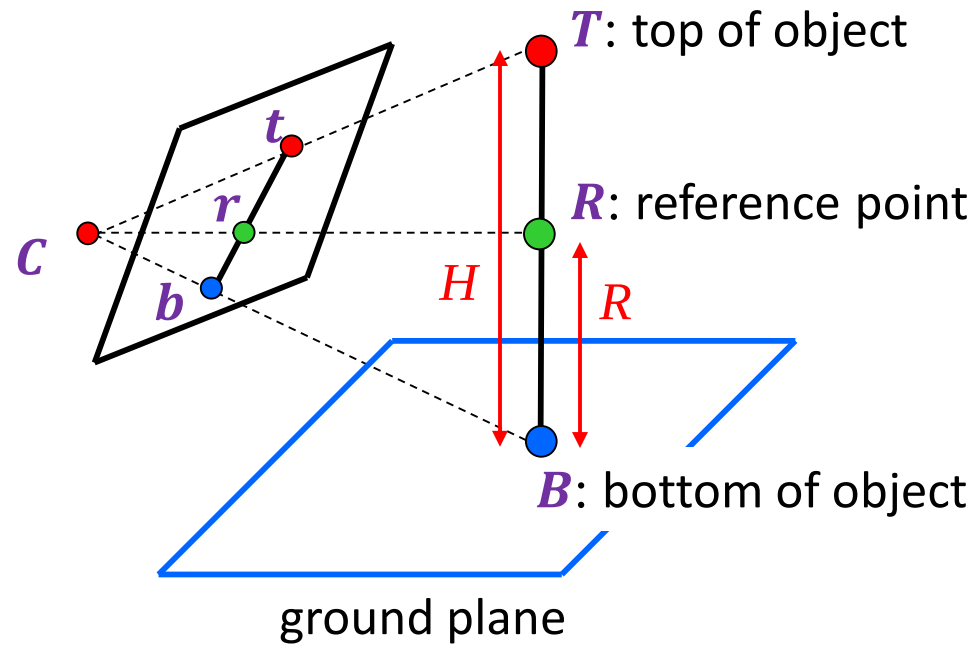


Measuring Height without a Ruler



- We can find $\frac{H}{R}$, but we need to do some extra work...

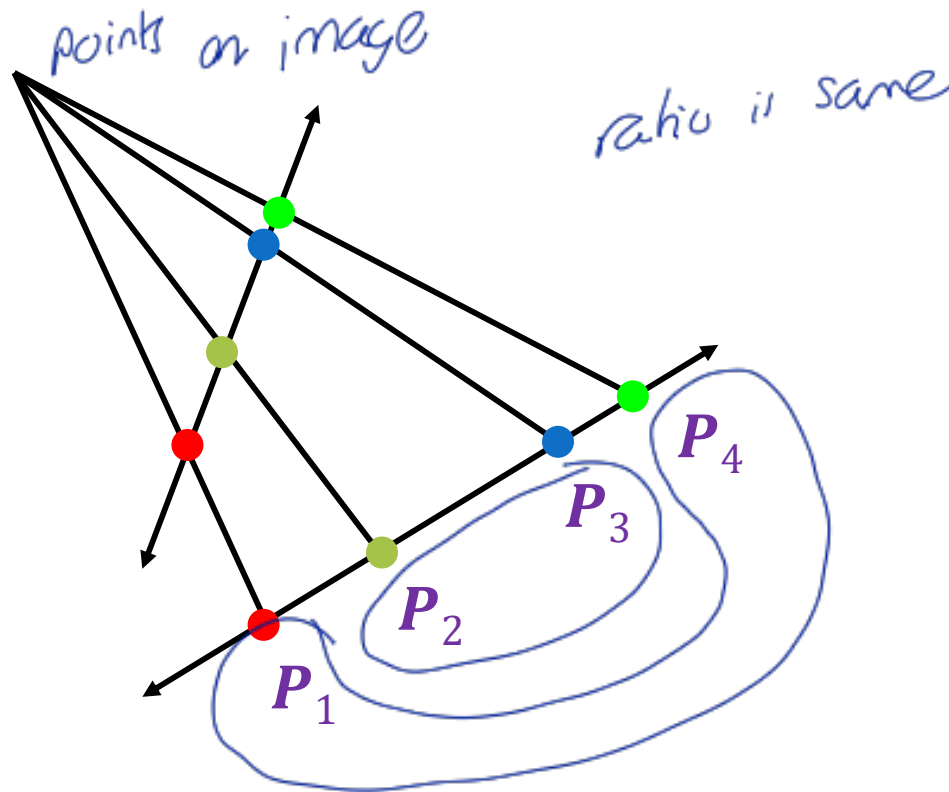
Measuring Height



To make 3D measurements based on image observations, we need an **invariant** that does not change under perspective projection from 3D to 2D

Cross-Ratio

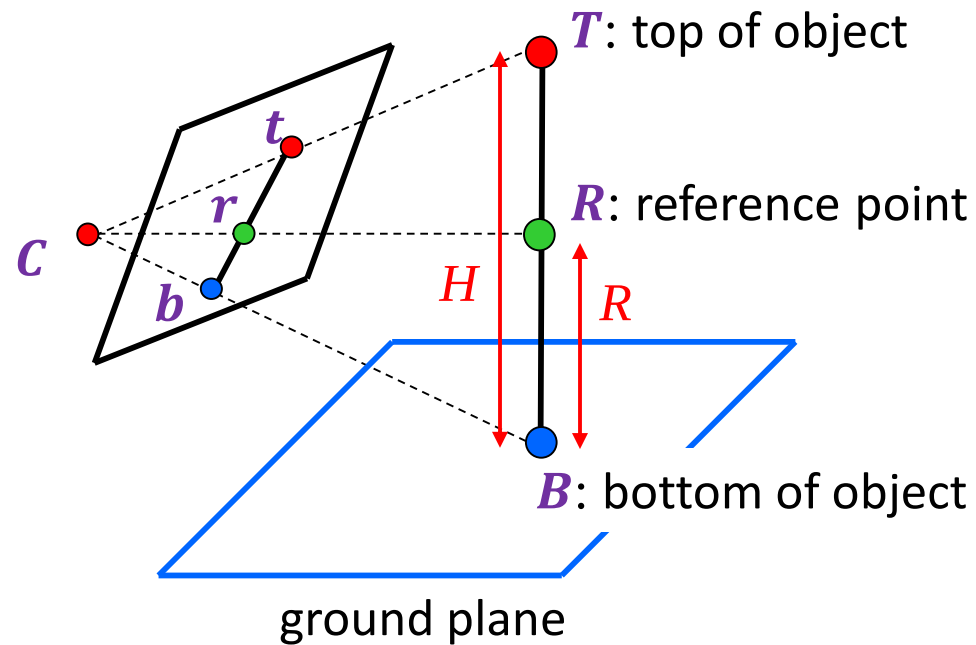
- Defined for four collinear points, invariant to projective transformations, including perspective projection from 3D to 2D
 - Also invariant to permutation of the points



$$\frac{\|P_3 - P_1\| \|P_4 - P_2\|}{\|P_3 - P_2\| \|P_4 - P_1\|}$$

Measuring Height

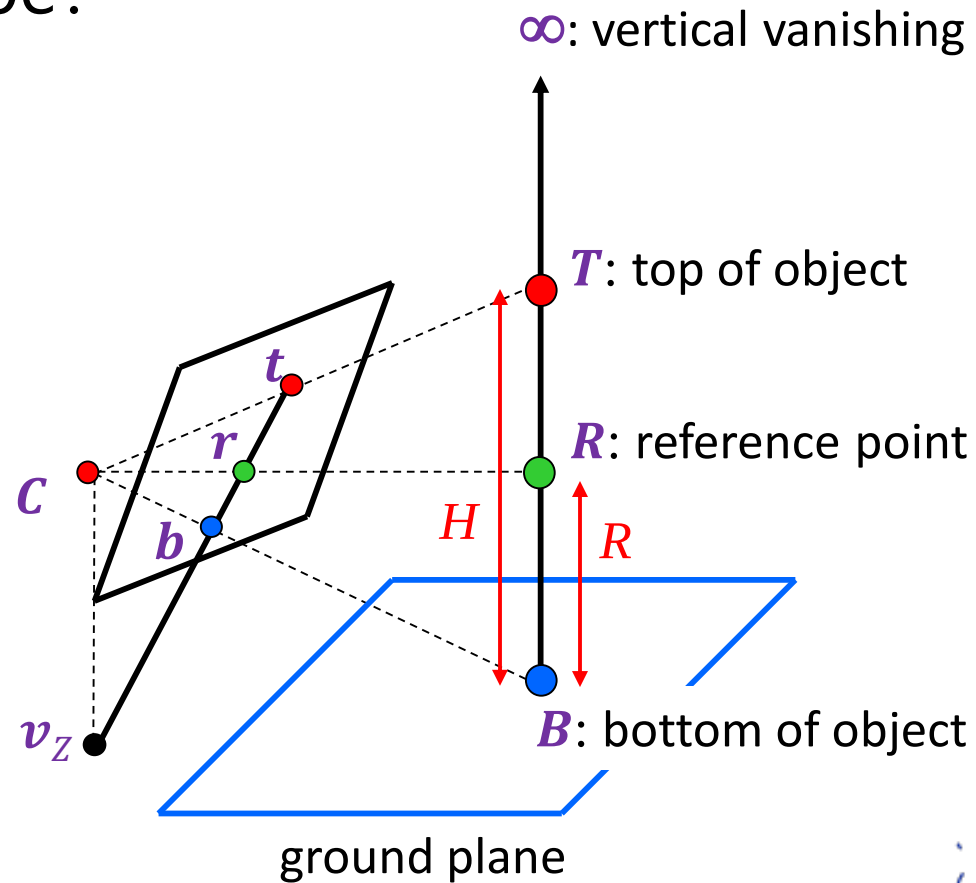
- We need a fourth point to compute a cross-ratio – what could it be?



$R \Rightarrow$ what we know
 $H \Rightarrow$ height in image

Measuring Height

- We need a fourth point to compute a cross-ratio – what could it be?



Scene cross ratio:

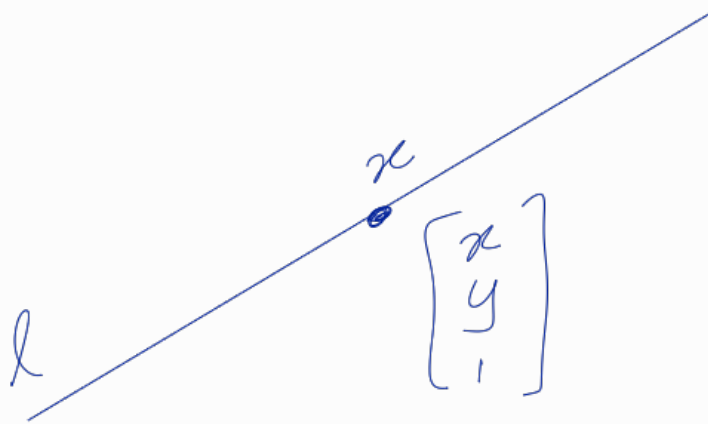
$$\frac{\|T - B\| \| \infty - R \|}{\|R - B\| \| \infty - T \|} = \frac{H}{R}$$

Image cross ratio:

$$\frac{\|t - b\| \|v_z - r\|}{\|r - b\| \|v_z - t\|} = \frac{h}{r}$$

if we know R we can get h

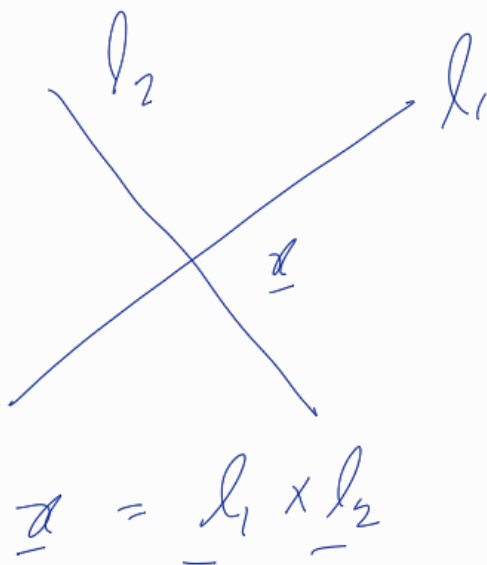
incidence $\Rightarrow$ point is on line



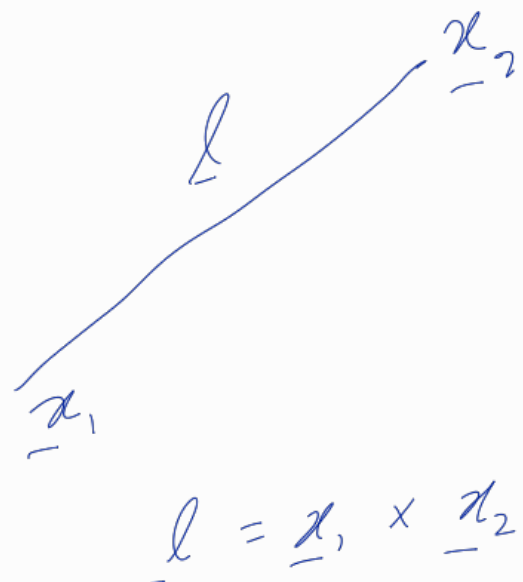
$$ax + by + c = 0$$

$$\begin{bmatrix} x & y & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0$$

line and point
cannot be
distinguished

$$\begin{cases} \underline{x}^T \underline{l} = 0 \\ \underline{l}^T \underline{x} = 0 \end{cases}$$


(2)



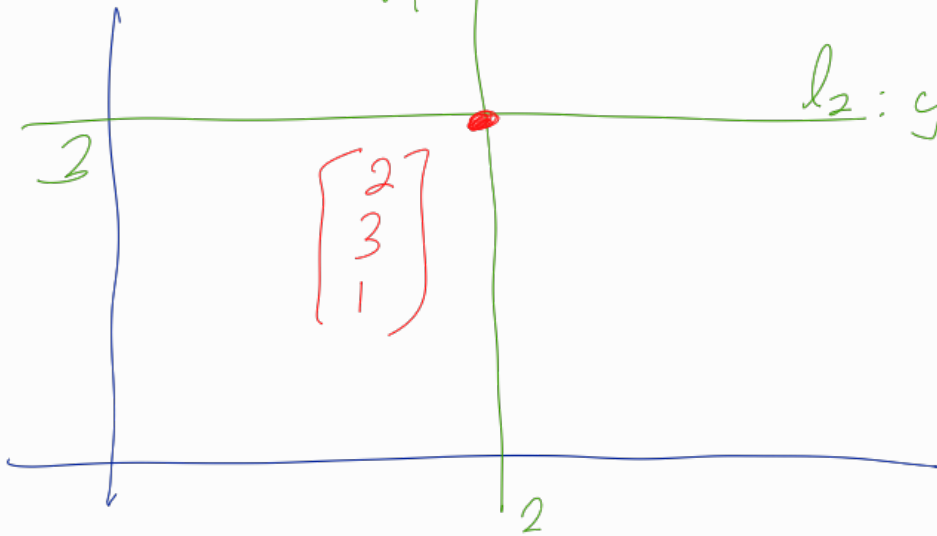
(3)

Proof

(2)

$$l_1: x=2$$

$$l_2: y=3$$



$$l_1: x-2=0$$

$$\underline{l}_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$$

$$l_2: y-3=0$$

$$\underline{l}_2 = \begin{bmatrix} 0 \\ 1 \\ -3 \end{bmatrix}$$

$$l_1 \times l_2 = \begin{vmatrix} i & j & k \\ 1 & 0 & -2 \\ 0 & 1 & -3 \end{vmatrix}$$

$$= +2i + 3j + k$$

$$= \underline{x}$$

How do we compute the coordinates of v and t ?

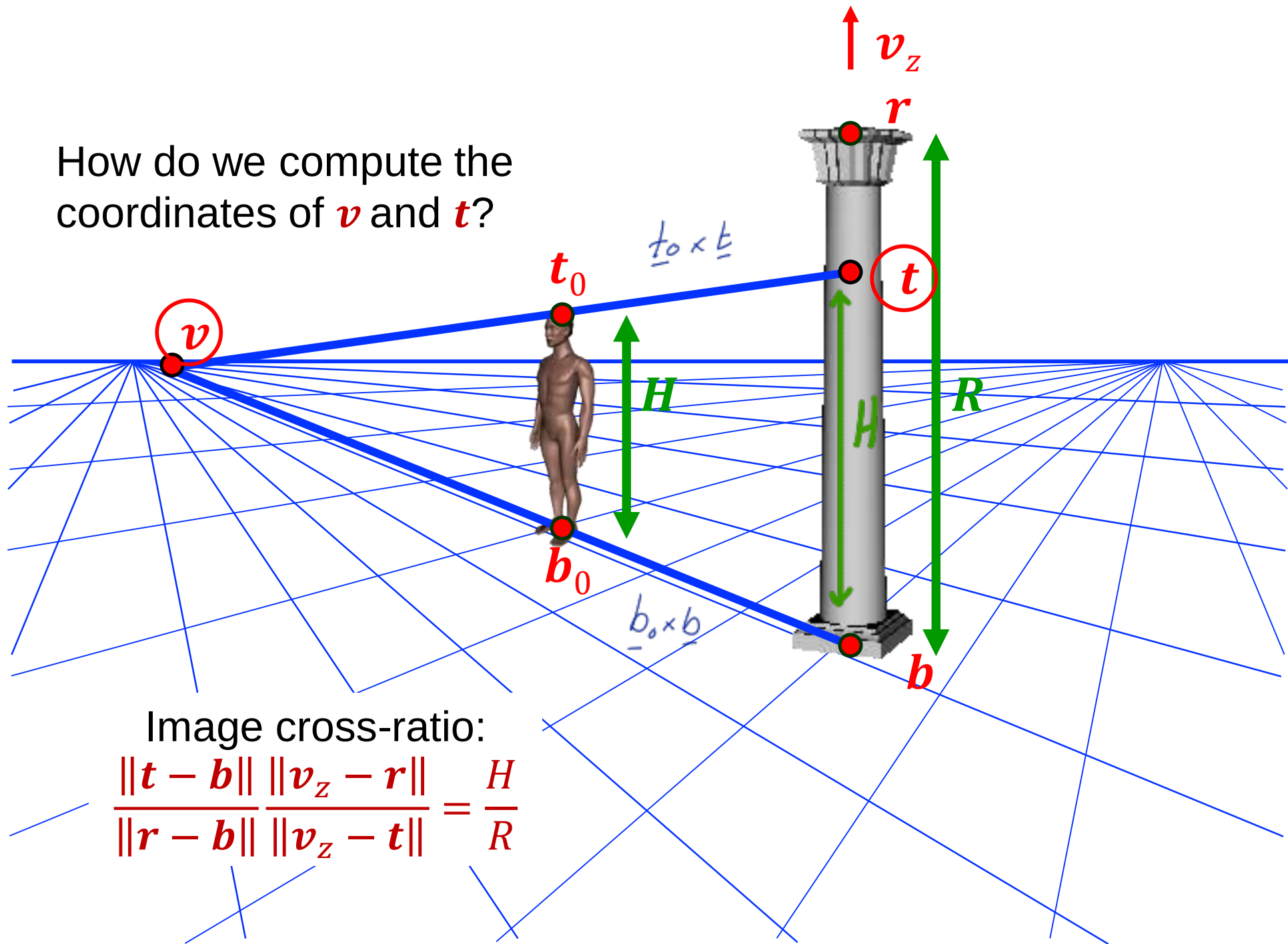


Image cross-ratio:

$$\frac{\|t - b\|}{\|r - b\|} \frac{\|v_z - r\|}{\|v_z - t\|} = \frac{H}{R}$$

2D Lines in Homogeneous Coordinates

- Line equation: $ax + by + c = 0$

$$\mathbf{l}^T \mathbf{x} = 0, \text{ where } \mathbf{l} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}, \mathbf{x} = \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}$$

- Line passing through two points: $\mathbf{l} = \mathbf{x}_1 \times \mathbf{x}_2$
- Intersection of two lines: $\mathbf{x} = \mathbf{l}_1 \times \mathbf{l}_2$
 - What is the intersection of two parallel lines?

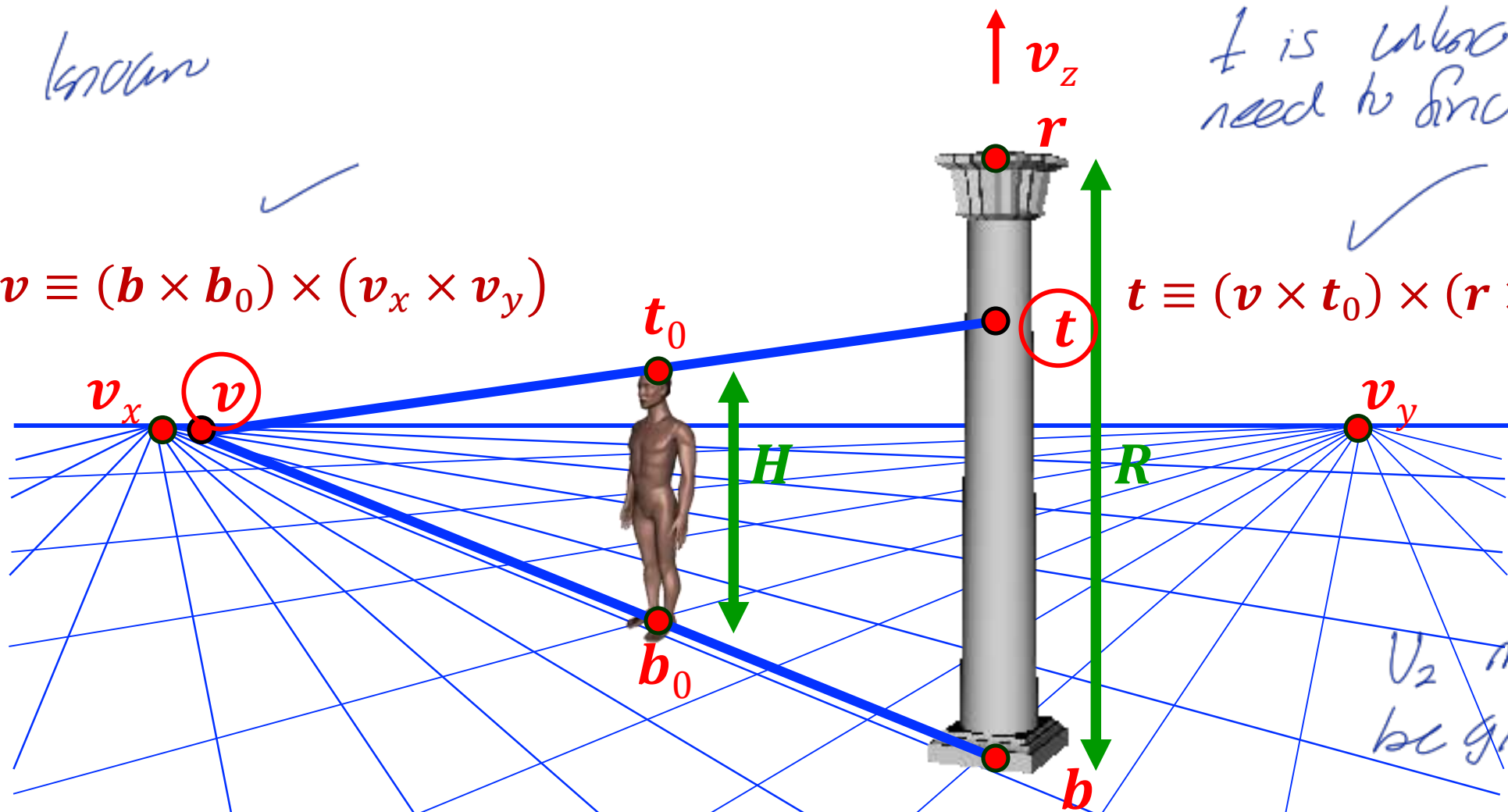
*Just
remember*

v_x, v_y known ✓

t is unknown
need to find. ✓

$$v \equiv (b \times b_0) \times (v_x \times v_y)$$

$$t \equiv (v \times t_0) \times (r \times b)$$



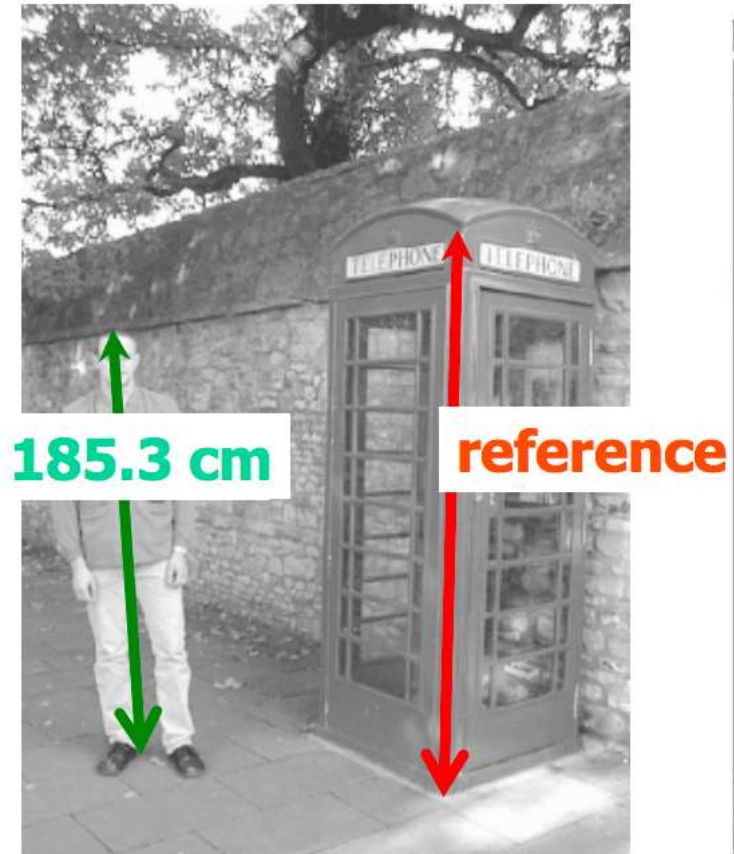
v_z must
be given

Image cross-ratio:

$$\frac{\|t - b\|}{\|r - b\|} \frac{\|v_z - r\|}{\|v_z - t\|} = \frac{H}{R}$$

if now all known
I can find H

Single-View Measurement Examples



A. Criminisi, I. Reid, and A. Zisserman, [Single View Metrology](#), IJCV 2000
Figure from [UPenn CIS580 slides](#)

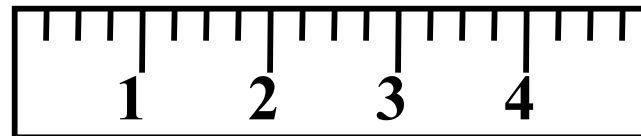
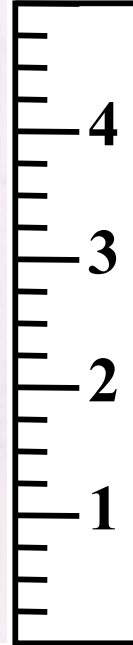
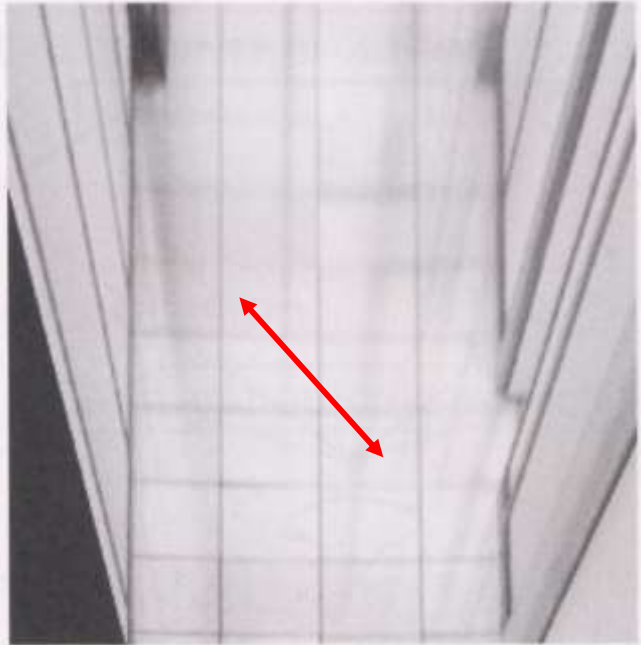
Another Example

- Are the heights of the two groups of people consistent with one another?
 - Measure heights using Christ as reference



Piero della Francesca, *Flagellation*, ca. 1455

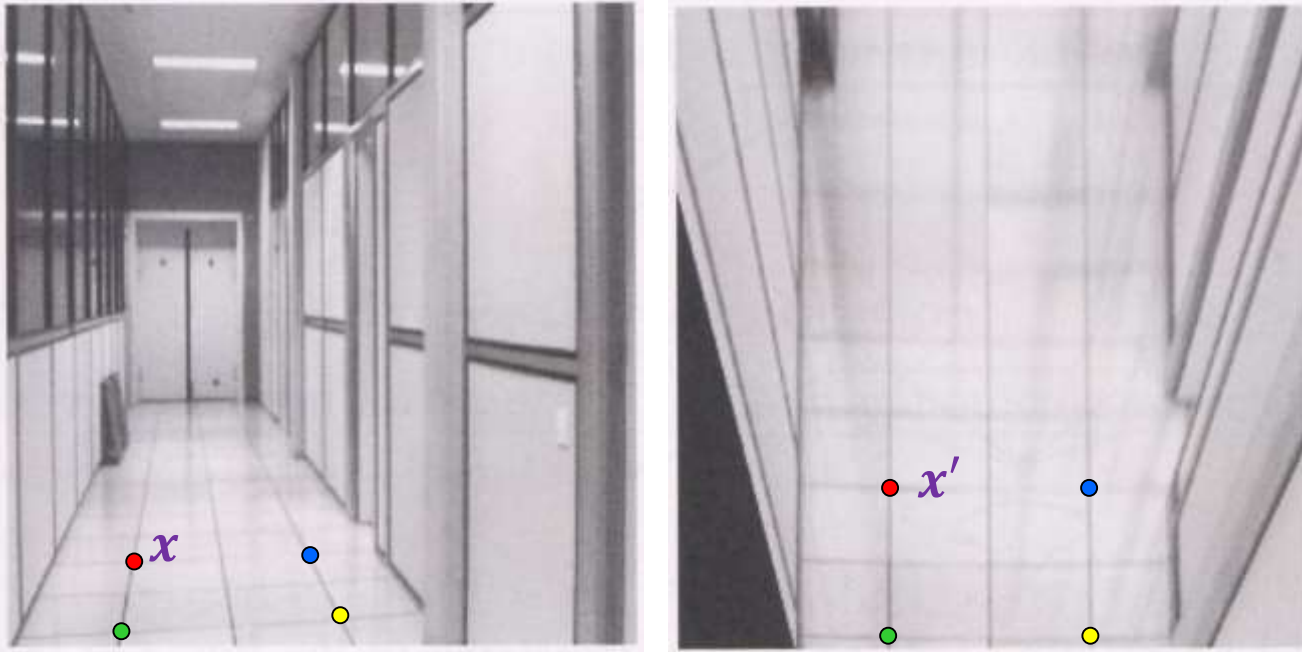
Measurements within Planes



huregaphiz
acsp

- Simplest approach: unwarp then measure
- What kind of warp is this?

Image Rectification

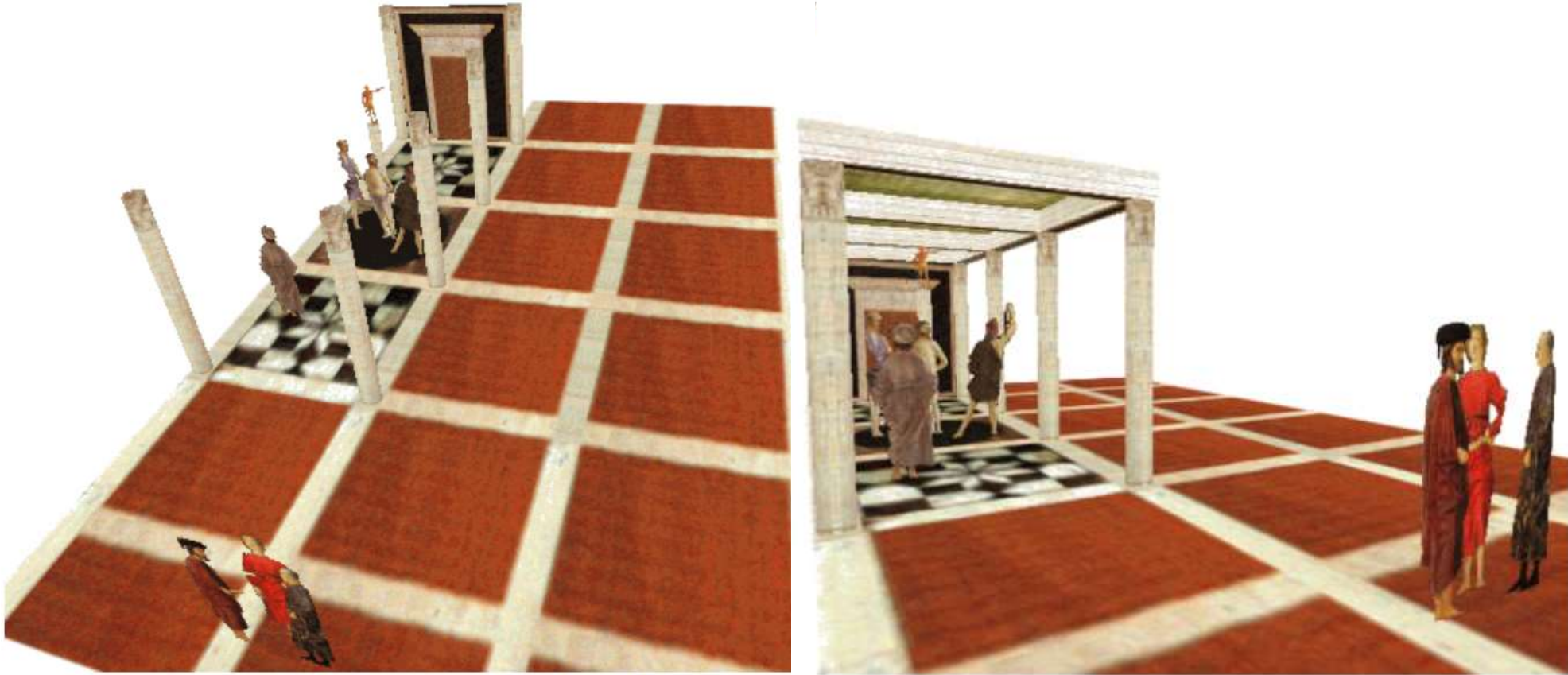


- To unwarp (rectify) an image, solve for homography H given four pairs of matches assumed to have a known configuration (e.g., square)

Image Rectification: Example



Putting Everything Together: Single-View Modeling

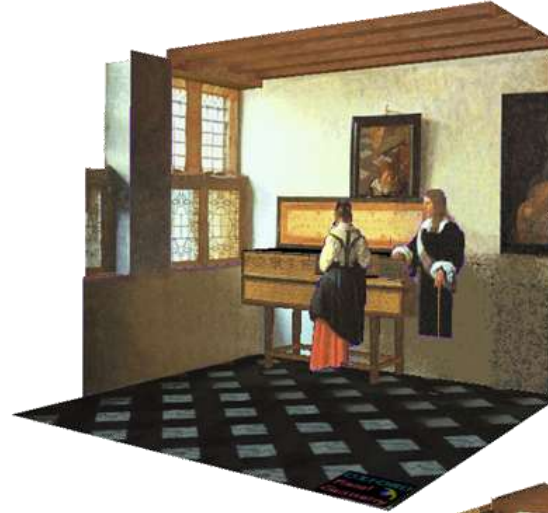


A. Criminisi et al. [Bringing Pictorial Space to Life: computer techniques for the analysis of paintings](#),
Proc. Computers and the History of Art, 2002

Putting Everything Together: Single-view modeling



J. Vermeer, *Music Lesson*, 1662

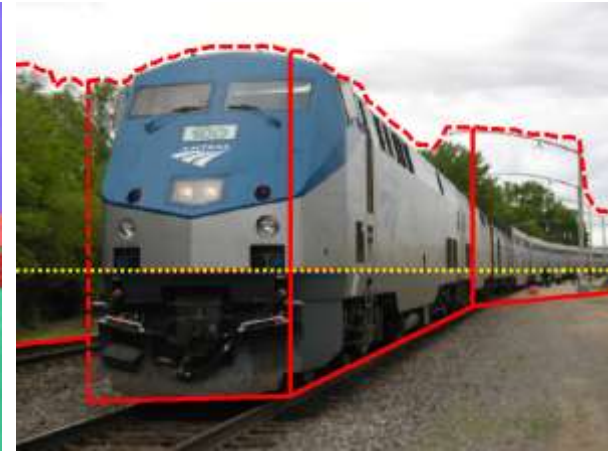
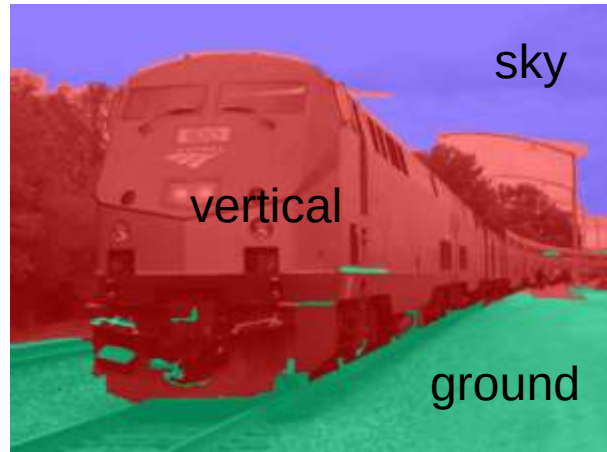


A. Criminisi et al. [Bringing Pictorial Space to Life: computer techniques for the analysis of paintings](#),
Proc. Computers and the History of Art, 2002

Outline

- Camera calibration using vanishing points
- Measurements from a single image
- Applications of single-view metrology

Application: Fully Automatic Modeling



Automatic Photo Pop-up

D. Hoiem A.A. Efros M. Hebert
Carnegie Mellon University

Application: Object Detection

unnecessary object detection removed
no objects above horizon.

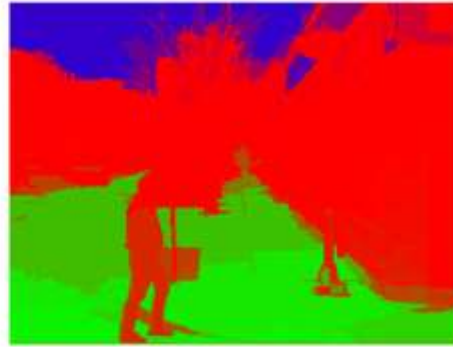


D. Hoiem, A. A. Efros, and M. Hebert. [Putting Objects in Perspective](#). CVPR 2006

Application: Object Detection



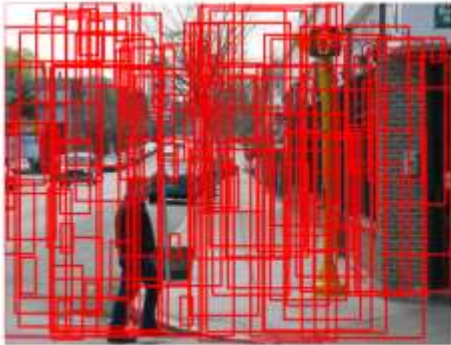
(a) input image



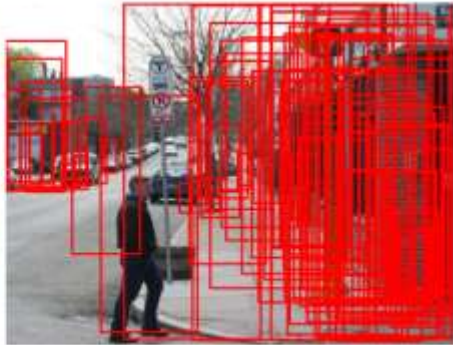
(c) surface orientation estimate



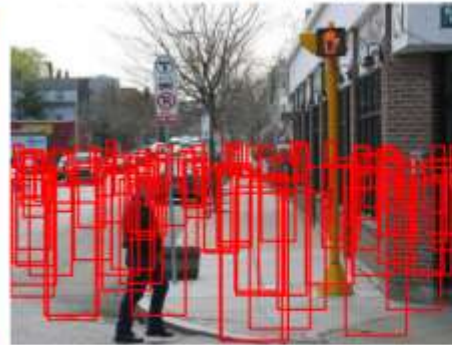
(e) $P(\text{viewpoint} \mid \text{objects})$



(b) $P(\text{person}) = \text{uniform}$



(d) $P(\text{person} \mid \text{geometry})$



(f) $P(\text{person} \mid \text{viewpoint})$



(g) $P(\text{person} \mid \text{viewpoint, geometry})$

D. Hoiem, A.A. Efros, and M. Hebert. [Putting Objects in Perspective](#). CVPR 2006

Application: Image Editing

- Inserting synthetic objects into images



Application: Image Editing

Inserting synthetic objects into images:

Video link:

<https://vimeo.com/28962540>